

Physical Action Effects in Dexterous Interaction Models: Resolution, Contact Acquisition, and Limits of Cold Replay

Exploratory working draft, revision2;1October2026

Abstract

Predicting recorded object trajectories and predicting the differences between actions are separate requirements for dexterous control. We investigate their measurement and learning in an existing Inspire-hand simulation substrate. Initial wrist pulses produce five-step position contrasts of 7.15 mm RMS against 0.10 mm observed zero-repeat contrasts, but fail a directional lifting screen. A contrast-supervised predictor also fails its actor-transfer screen. Current-geometry auditing reveals that a reference force-proxy trigger included an entire test motion with hand-object gaps of 55-105 mm. On larger archived data, a predicted-hand-motion factorization improves near one-step error by 8.33%, failing its fixed 10% gate; its simpler nominal-PD-motion control improves 11.18%, motivating a fresh intervention test. New geometry-conditioned acquisition finds 384 interacting-proximity windows, but a cold-replay matching precondition fails before treatment, preventing the intended causal model comparison. An independent exact-null experiment does not support inverse-recovery-induced prediction distortion. These exploratory results identify useful acquisition and measurement constraints. They establish neither a new general modeling method nor a manipulation-policy benefit.

1. Problem and relation to prior work

An interaction model can fit common motion while failing to compare alternative actions. Useful comparisons additionally require experiments capable of measuring physical differences at the states in which actions are selected. We examine this prerequisite before committing further resources to a policy. The repository provides self-trained actors, geometric models and a DExplore simulation substrate[xu2025]. Historical policy audits are motivation rather than fresh evidence for this manuscript.

Shared particle actions and cross-embodiment world models already support planning [he2025]. Value-aware objectives[farahmand2017], predictor-level action recovery[qi2026] and multi-step controllability representations[rudolph2024] are established. Contact-consistent replay and actuator residuals are also developed by Contact-UAN[fey2026]. Our small physical MLP studies do not reproduce these systems or establish priority for their general principles. The present contribution is an exploratory evidence record with frozen hypotheses and retained failures.

2. Initial physical and contrast-learning screens

Four panels combine actor seeds 286/287 with environment seeds 288/289, each with 32 environments for each of three airplane motions. A retrospective trigger schedules one step after the first reference event where independent hand and object force proxies are active. A one-step wrist-z action change requests a plus or minus 10-mm PD target offset. This is a desired offset, not measured wrist travel. All subsequent normalized actions and RNG schedules are replayed. Two zero arms measure repeated process variation. Each object response subtracts its own actual pre-intervention position. Pulse and repeat contrasts are summarized by vector RMS.

Steps	Pulse RMS (mm)	Repeat RMS (mm)
1	1.914	0.000
2	2.572	0.084
5	7.150	0.101
10	18.827	0.444

Steps	Pulse RMS (mm)	Repeat RMS (mm)
20	75.463	6.081
30	105.792	9.638

Table1. Initial fixed panels: pulse versus observed zero-repeat position contrasts.

At five and ten steps, contrast ratios are 70.44 and 42.36. At one step, the observed zero repeat is exactly zero; its ratio is undefined rather than made finite by an arbitrary denominator. Mixed vertical signs fail the joint physical gate. No beneficial lifting or population confidence bound follows from a large observed contrast.

The subsequent vector-learning decision retains this failure. It fits 192 actor-286 windows and tests 192 actor-287 windows using four past/current physical states and ten preset replay actions. Future observed physical states are absent from inputs. Matched two-layer 128-unit SiLU networks predict five/ten-step position responses; three seeds each receive 1000 Adam updates. Differential training adds measured plus-minus contrast loss. All normalization uses fit rows. Fixed test panels were already inspected by the physical experiment, making this reused-data exploration.

Input/objective	Five steps (mm)	Ten steps (mm)
Factual	10.121	26.546
Differential	10.114	26.990
No pulse	9.922	26.456
Global fit mean	10.061	26.259
Motion fit mean (privileged)	10.137	26.340

Table2. Actor-transfer contrast RMSE, millimeters; learned rows average three optimization seeds.

Differential training does not improve over factual or global fit-mean controls. Vertical sign agreement is 32.6%/31.1%, below its fixed 75% gate. Both initial screens are UNPROMISING. Increasing steps or choosing another seed is not used to salvage these exact designs. Physical acquisition took 481.9 seconds; fitting took 17.2 seconds on one RTX 3090.

3. Geometry and larger archived predictive screen

Full sampled current geometry places 318 of 384 old triggers within 20 mm. However, actor 287's motion 1 is entirely outside that range in both old test panels, with gaps 55-105 mm. Independent force proxies can both activate due to table contacts. The geometric shift is documented, without identifying it as the sole cause of learning failure or redefining the old gate.

We next use two immutable, episode-disjoint historical collections, 960 complete episodes each. Nonterminal frames at step modulo 16 equal to zero yield 32902 fit and 32896 test rows. These collections have been used by previous research and are not pristine Validation data. The current-near test stratum contains 1756 frames from 639 episodes; proximity uses a frozen stride-8 surface convention.

The target is one-step object-position residual over constant velocity, expressed in the current object frame. State input includes joint angles and velocities, object velocities, gravity direction, object height, palm/tip relative positions and force proxies. Matched effect networks receive either zero action slots, native commands, exact PD-target/FK palm-tip motion, predicted realized hand motion, or measured future hand motion. The last is a privileged diagnostic. An independently fit actuator model predicts joint residuals before the learned-motion variant's FK. Effect networks have two 128-unit SiLU layers, fixed 1000 updates and three seeds. Sampling is half-near and half-all fit rows. No test-selected checkpoint is used.

Effect input	Near RMSE (mm)	All RMSE (mm)
State only	17.180	7.328
Native command	16.436	7.330
Nominal PD motion	14.599	7.093
Predicted hand motion	15.066	7.206
Measured future hand (privileged)	13.994	7.001
Passive	23.400	7.587

Table3. Archived one-step vector prediction error, seed-mean millimeters. Near and all-state strata differ.

The predicted-motion primary method gains 8.33% over command input, failing the 10% requirement despite beating state-only, passive and its geometric transport control. Its gate remains UNPROMISING. Actuator wrist-position errors are 11.75-12.07 mm near the object. Measured future hand motion includes possible contact feedback and cannot establish a deployable causal mediator.

The separately reported nominal-motion variant gains 11.18% and is better in each seed. This motivates a separately frozen fresh causal test; it does not retroactively replace the primary method. The label-free nominal no-slip transport control is inaccurate, with near RMSE 133.64 mm. The full screen took 53.64 seconds. All 30 saved held RMSE metrics were independently recomputed.

4. Fresh proximity acquisition and failed matching requirement

New environment seeds 490/491 are combined with the same two actors. A compact reference records up to 360 steps, stopping at the first native termination. For each environment, acquisition selects the first pre-action step at or after step 10 with five remaining nonterminal transitions, both current force proxies, and full sampled surface distance at most 20 mm. Future object outcomes never choose a trigger. Frozen assets provide 1538 hand and 1024 object points.

All 384 environments acquire a qualifying state, including all three motions in every panel. The maximum sampled gap per panel is 16.60,15.57,18.83,18.12 mm. Eight cold processes per panel implement two zeros and one-step plus/minus pulses in world x,y,z. Initial tensors, physical properties, actor/RMS states, actions and RNG schedules are checked. Warm solver state is not asserted to be cloneable. The existing effect models are frozen before acquisition; no refit is allowed.

The intended comparison requires pre-intervention position RMS at most 0.05 mm, maximum joint difference at most 0.0001, and maximum quaternion-component difference at most 0.0001. Joint coordinates combine translations in meters and angles in radians; quaternion differences are dimensionless component differences.

Panel	Position RMS (mm)	Joint max	Quaternion max
t286_s490	0.003554	0.0000732	0.0000670
t286_s491	0.010859	0.0002204	0.0006188
t287_s490	0.007110	0.0000000	0.0000416
t287_s491	0.013398	0.0000982	0.0000881

Table4. Maximum pre-intervention mismatch over arms relative to zero-a. Limits: position 0.05 mm; joint and quaternion components 0.0001.

All 32 physical arms complete, but seven arms of panel actor286/seed491 fail the joint/orientation precondition. The first detected failure is zero-b, before any action intervention. Execution stops with FAILED; the scientific transfer screen is UNCLEAR due to invalid matching. No favorable arm or window is retained, no threshold is widened, and no

pooled causal score or model winner is reported. Acquisition and the failed analysis took 1152.21 seconds. This is a measurement blocker, not a refutation of the nominal predictor.

5. Exact-null recovery mechanism screen

Six native commands are overwritten by dependent finger targets. Randomizing them leaves every PD target exactly unchanged; all 32896 held rows verify this bitwise. We test whether recovering such independent null commands from model-generated object predictions induces false action responses. A truthful physical transition cannot recover an independent normalized null coordinate with population MSE below one. This is an elementary conditional-variance bound, not a novel theorem. Mutual-information sufficiency limits are already studied[rakelly2021].

Every minibatch independently randomizes null commands while keeping physical labels fixed. Four matched physical predictors receive factual loss only, full inverse loss, effective-command inverse loss, or hard-canonicalized forward input with effective inverse loss. Inverse heads take current state and generated object residuals, never actual future state. Seeds 511-513 each train for 1000 updates; the auxiliary coefficient is fixed at one. Hard canonicalization guarantees null invariance by construction, without establishing learned causal correctness.

Variant	Near RMSE (mm)	Null response (mm)	Null inverse MSE
Factual only	16.557	1.118	not trained
Full inverse	16.534	1.079	1.002
Effective inverse	16.531	1.069	1.023
Hard quotient + inverse	16.564	0.000	1.023

Table5. Exact-null intervention diagnostics, near stratum and seed means. Untrained null-head coordinates in effective/quotient variants are diagnostic only.

Full inverse null error remains near the population floor and its null response is below factual-only in every seed. Three of five interference gates fail: UNPROMISING. The proposed inverse-induced null-encoding mechanism is not supported here. Hard canonicalization removes null response, with little factual-error change. This screen took 33.06 seconds; 66 saved prediction/recovery metrics were independently audited. It is neither an AD-WM reproduction nor a general rejection of inverse learning.

6. Limitations and research decision

These experiments involve one object, three shared motions, two existing actors and small fixed fitting procedures. Contact proxies plus sampled proximity are not attributed pairwise forces. Short-horizon measurement is not stable lift, retention or manipulation utility. No policy training, unseen-object/hand evaluation or hardware deployment is presented. Optimization seeds do not create independent physical datasets. Existing papers' claims are not replaced by these diagnostic tests.

The fresh matching failure should not be repaired by tuning a threshold on its observed outcomes. Randomized intervention assignment can instead identify effect-risk differences without claiming individually cloned warm states. Such a trial would require new states, assignment independent of pretreatment history, known propensities and a frozen estimand. Retrospective selection among the failed cold arms is insufficient. This is the next measurement decision, not an achieved method or journal-level claim.

The original user objective remains a journal-level research outcome. This working paper does not meet it: a distinctive method and consequential matched task evidence remain necessary. Negative outcomes and the unresolved causal screen are retained to guide further work rather than converted into positive conclusions.

Reproducibility

All new work is isolated in branch agent/contact-response-cm. Original checkpoints, motion assets and prior evidence are read-only. Physical/differential code is fixed at466e80d/44f22ea; factorization atbbd5ba4; fresh collection at59de1cf; null recovery at33d1085. Manifests preserve commands, input SHA, GPU UUID, budgets and failures. Every table is generated directly from recorded result JSON, including the failed fresh matching audit. This PDF is a ReportLab review copy; portable LaTeX source and tables are also retained. The first manuscript revision is preserved separately.

References

- [he2025] Z.He et al. Scaling Cross-Embodiment World Models for Dexterous Manipulation. arXiv:2511.01177,2025. <https://arxiv.org/abs/2511.01177>.
- [xu2025] S.Xu et al. Dexplore:Scalable Neural Control for Dexterous Manipulation from Reference-Scoped Exploration.2025. <https://arxiv.org/abs/2509.09671>.
- [farahmand2017] A.-M.Farahmand,A.Barreto,D.Nikovski. Value-Aware Loss Function for Model-based Reinforcement Learning.2017. <https://proceedings.mlr.press/v54/farahmand17a.html>.
- [qiu2026] J.Qiu et al. AD-WM:Action-Discriminative World Models for Counterfactual Model Predictive Control.2026. <https://arxiv.org/abs/2609.30264v2>.
- [rudolph2024] M.Rudolph et al. Learning Action-based Representations Using Invariance. 2024. <https://arxiv.org/abs/2403.16369>.
- [fey2026] N.Fey et al. Contact-UAN:Unsupervised Actuator Networks as a Residual World Model for Humanoid Control.2026. <https://contact-uan.csail.mit.edu/>.
- [rakelly2021] K.Rakelly,A.Gupta,C.Florensa,S.Levine. Which Mutual-Information Representation Learning Objectives are Sufficient for Control?2021. <https://arxiv.org/abs/2106.07278>.